

ORIGINAL RESEARCH

1-Year Outcomes of Drug-Coated Balloon Angioplasty Versus Drug-Eluting Stent for Coronary Artery Bypass Graft Lesions

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ABSTRACT

BACKGROUND Graft failure after coronary artery bypass grafting is a common clinical problem associated with adverse cardiovascular outcomes. Although percutaneous coronary intervention (PCI) for bypass graft lesions is frequently performed, the optimal device strategy remains debated. Recently, drug-coated balloon (DCB) has emerged as a stentless alternative.

OBJECTIVES This study aimed to compare 1-year clinical outcomes between drug-eluting stent (DES) and DCB-only strategies in patients undergoing PCI for bypass graft lesions.

METHODS We analyzed data from the J-PCI OUTCOME (Japanese Percutaneous Coronary Intervention Outcome Registry), a nationwide registry in Japan. Patients who underwent PCI with DES or stentless PCI with DCB for bypass graft lesions between 2017 and 2020 were included. The primary endpoint was major adverse cardiovascular events (MACEs), defined as a composite of all-cause death, acute coronary syndrome, and stroke within 1 year after PCI. Secondary endpoints included individual components of MACE, major bleeding, and planned repeat revascularization.

RESULTS Of the 610 PCIs for bypass graft lesions, 184 (30.2%) were stentless PCIs with DCB. One-year MACE rates did not differ significantly between the DES and DCB-only strategies (10.3% vs 8.6%; $P = 0.69$). No statistically significant differences were observed in the secondary endpoints. After multivariable adjustment, the DCB-only strategy was not associated with a significant difference in MACE risk after PCI compared with the DES-based strategy (adjusted HR: 1.01; 95% CI: 0.58-1.76; $P = 0.96$).

CONCLUSIONS Stentless PCI with DCB was frequently used in patients undergoing PCI for bypass graft lesions in Japan and was not associated with a significant difference in 1-year clinical outcomes compared with contemporary DES-based PCI strategies. (JACC Cardiovasc Interv. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

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ABBREVIATIONS
AND ACRONYMS**ACS** = acute coronary syndrome**AG** = arterial graft**CABG** = coronary artery bypass grafting**DCB** = drug-coated balloon**DES** = drug-eluting stent(s)**MACE** = major adverse cardiovascular event(s)**NCD** = National Clinical Database**PCI** = percutaneous coronary intervention**SVG** = saphenous vein graft

Graft failure after coronary artery bypass grafting (CABG) is a significant clinical problem that contributes to late morbidity and the need for additional revascularization procedures.¹ Despite substantial advancements in coronary revascularization strategies, there is limited contemporary real-world data on the optimal revascularization strategies for this patient population.²⁻⁴ The target vessel for percutaneous coronary intervention (PCI) in patients with graft failure depends on the clinical situation and coronary anatomy.⁴⁻⁶ Current guidelines recommend PCI to a native vessel over PCI to a severely diseased saphenous vein graft (SVG) when anatomically feasible.⁶ However, the randomized PROCTOR (PeRcutaneous cORonary intervention of native Coronary arTery versus venous bypass graft in patients with prior cORonary artery bypass graft surgery) trial reported that SVG PCI was associated with improved 1-year clinical outcomes compared with native vessel PCI.² Native vessel PCI after CABG often involves chronic total occlusions or complex lesions, which may increase procedural difficulty and procedural risk.^{4,5} Consequently, PCI to a diseased bypass graft remains common in clinical practice.

Another important procedural question is whether outcomes after PCI for bypass graft lesions are optimized by implanting drug-eluting stents (DESs) with meticulous lesion preparation, full stent expansion, and apposition or by using a stentless strategy with drug-coated balloons (DCBs) alone. Because aggressive stent expansion in SVG may increase the risk of distal embolization with slow-flow/no-reflow phenomena, graft dissection, and perforation,^{4,5} stentless PCI with DCB has become an attractive alternative in certain cases.^{7,8} However, pivotal randomized trials of DCB-based PCI excluded patients with previous CABG or those undergoing PCI for bypass graft lesions.^{9,10} A small study that compared PCIs with DCB and DES for de novo SVG lesions reported similar rates of major adverse cardiovascular events (MACEs) and repeat revascularization.⁷ Furthermore, clinical outcomes may differ between PCIs to SVG and arterial graft

(AG) because AGs often have small reference vessel diameters, frequently ≤ 3.0 mm,^{11,12} which may further limit the suitability of DES implantation.

To address this knowledge gap and provide real-world data in this high-risk population, we compared 1-year clinical outcomes between DES and DCB-only strategies in patients undergoing PCI for bypass graft lesions using data from a large nationwide registry.

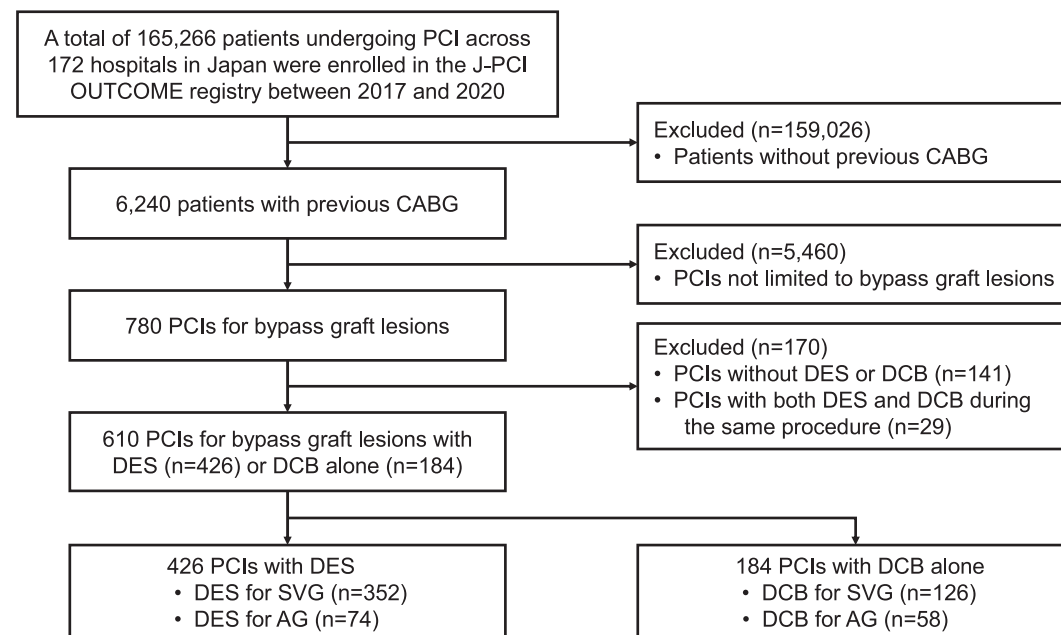
METHODS

STUDY POPULATION. This was a retrospective observational analysis using data from the J-PCI OUTCOME (Japanese Percutaneous Coronary Intervention Outcome Registry), a prospective, nationwide, multicenter registry established and endorsed by the Japanese Association of Cardiovascular Intervention and Therapeutics. The J-PCI OUTCOME registry collected standardized 1-year postdischarge clinical outcomes for patients undergoing PCI at 172 hospitals between January 2017 and December 2020.¹³⁻¹⁵ Participating hospitals met the selection criterion of performing more than 200 PCIs annually. Outcomes were recorded in a secure, web-based electronic data capture system managed by the National Clinical Database (NCD).

The registry's design, data collection process, and variable definitions have been described previously.¹³⁻¹⁸ Briefly, clinical, procedural, and in-hospital outcome data were prospectively collected using an electronic data capture system (ie, the NCD of Japan) with standardized definitions for all variables.¹⁶ In-hospital outcomes were recorded at discharge or death. Patients who died during hospitalization after PCI were excluded. The J-PCI OUTCOME registry focuses on patient-level clinical outcomes after hospital discharge.¹⁶ Accordingly, lesion-specific endpoints, such as target lesion failure, in-stent restenosis, and lesion-level revascularization were not included in the dataset. Follow-ups at 3 months and 1 year were conducted via telephone interview or medical record review. Site data managers reviewed the outcome data, and the J-PCI program uses structured processes, including audits, to ensure data quality.

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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FIGURE 1 Flow Diagram Illustrating Patient Selection From the J-PCI OUTCOME Registry

This figure illustrates the stepwise selection of the study population from the nationwide J-PCI OUTCOME (Japanese Percutaneous Coronary Intervention Outcome Registry). First, among all patients undergoing percutaneous coronary intervention (PCI) between 2017 and 2020, those with previous coronary artery bypass grafting (CABG) were identified. Then, cases not involving bypass graft lesions or treatment with drug-eluting stents (DESs) or drug-coated balloons (DCBs) were excluded. Cases treated with both DES and DCB were excluded because of an inability to distinguish between planned hybrid and bailout strategies. The final cohort included 610 PCI procedures, which were used to compare clinical outcomes between treatment strategies. AG = arterial graft; SVG = saphenous vein graft.

The J-PCI OUTCOME registry protocol, which incorporated an opt-out consent process, was approved by an independent central ethics committee at the Japan Clinical Research Promotion Network in Osaka, Japan, as well as by the institutional review boards of each participating institution. The study was conducted in accordance with the Declaration of Helsinki. Because of the retrospective and observational nature of the study, written informed consent was waived.

This analysis focused on patients who underwent PCI with DES or stentless PCI with DCB for bypass graft lesions. The DES group was not intended to represent a platform-specific control group; rather, it represented contemporary real-world PCI practices with DESs in Japan during the study period. Of the 165,266 patients who underwent PCI during the study period, 6,240 patients with a history of CABG were identified after excluding those without previous CABG ($n = 159,026$). Next, we excluded 5,460 PCIs that did not involve bypass graft vessels, 141 PCIs that were performed without DES or DCB, and 29

PCIs in which both DES and DCB were used, because the registry could not reliably distinguish between planned hybrid strategies and bailout DES implantation after an initially intended DCB-only strategy. The final study cohort consisted of 610 PCIs, including 426 PCIs with DES (352 SVGs and 74 AGs) and 184 stentless PCIs with DCB (126 SVGs and 58 AGs) (Figure 1). During the study period, the paclitaxel-coated SeQuent Please balloon (B. Braun) was the only DCB available in Japan for clinical use.

CLINICAL OUTCOMES. The primary endpoint was MACE, defined as a composite of all-cause death, nonfatal acute coronary syndrome (ACS), and nonfatal stroke within 1 year after PCI. The secondary endpoints were all-cause mortality, cardiovascular death (defined as death due to ACS, sudden cardiac death, heart failure, stroke, procedure-related complications, or major bleeding), noncardiovascular death, nonfatal ACS, nonfatal stroke, nonfatal major bleeding, and planned repeat revascularization. Time to event was defined as the interval from hospital

discharge to the occurrence of an outcome event or completion of the 1-year follow-up, whichever occurred first. The clinical endpoints were defined according to the 2017 Cardiovascular and Stroke Endpoint Definitions for Clinical Trials,¹⁹ as adopted by the J-PCI OUTCOME registry.

STATISTICAL ANALYSIS. Baseline clinical and procedural characteristics were summarized and compared between the DES and DCB groups. Continuous variables are presented as mean $\pm$ SD, and categorical variables are presented as counts and percentages. Student's *t*-test was used to compare continuous variables between the groups, and the chi-square test was used for categorical variables. Cumulative incidences of clinical outcomes were estimated using the Kaplan-Meier method, and group differences were assessed using the log-rank test. Cox proportional hazards regression models were used to estimate HRs and 95% CIs for comparisons between the DES and DCB groups. The proportional hazards assumption was verified by visually inspecting log-log plots and testing Schoenfeld residuals. Multivariable models included the treatment group (DES vs DCB) and were adjusted for prespecified, clinically relevant covariates, including age, sex, chronic kidney disease (defined as the presence of proteinuria, serum creatinine ≥ 1.3 mg/dL, or estimated glomerular filtration rate ≤ 60 mL/min/1.73 m²),¹⁶ and presentation with ACS, based on clinical relevance¹⁷ and the anticipated number of primary outcome events (approximately 1 variable per 10 events) to minimize overfitting. These variables were considered potential common causes of both treatment selection and outcomes, analogous to confounders identified in a directed acyclic graph framework. Multicollinearity among the covariates was assessed and confirmed to be insignificant (ie, the variance inflation factors were all less than 5). Subgroup analyses were performed according to PCI target vessel (SVG or AG) to explore potential heterogeneity in treatment effects. A 2-sided $P < 0.05$ was considered statistically significant. All statistical analyses were performed using R software, version 4.5.2 (R Foundation for Statistical Computing).

RESULTS

Of the 610 patients undergoing PCI for bypass graft lesions, the mean age was 72.8 ± 10.4 years, 500 (82.0%) were men, 301 (49.8%) had a previous history of myocardial infarction, and 207 (34.1%) underwent PCI for ACS (Table 1). Overall, baseline patient characteristics were similar between the DES

and DCB groups. However, the DES group had a numerically higher prevalence of ACS than the DCB group (36.2% vs 29.1%; $P = 0.051$). There were no cases of in-hospital death, emergency surgery, or stent thrombosis during hospitalization.

1-YEAR OUTCOMES AFTER PCI. Clinical outcomes at 1 year after PCI for bypass graft lesions are shown in Table 2. MACEs occurred in 41 patients (10.3%) in the DES group and 15 patients (8.6%) in the DCB group. There was no significant difference in MACE rates after PCI between the 2 groups (HR: 0.90; 95% CI: 0.52-1.55; $P = 0.69$; Figure 2). The results were consistent when stratified by SVG and AG subgroups. Similarly, there were no significant differences in all-cause death, cardiovascular death, nonfatal ACS, nonfatal stroke, nonfatal major bleeding, or planned repeat revascularization between the DES and DCB groups (Figure 3).

The HRs for MACE incidence at 1 year after PCI in multivariable Cox models are presented in Table 3. The DCB-only strategy was not associated with a significant difference in MACE risk after PCI compared with the DES-based strategy (adjusted HR: 1.01; 95% CI: 0.58-1.76; $P = 0.96$). The global test of the multivariable model confirmed that the proportional hazards assumption was not violated ($P = 0.24$). This nonsignificant difference was consistent across SVG and AG subgroups. Chronic kidney disease (adjusted HR: 2.07; 95% CI: 1.24-3.48; $P = 0.006$) and ACS at presentation (adjusted HR: 2.16; 95% CI: 1.30-3.60; $P = 0.003$) were significantly associated with an increased risk of MACE.

DISCUSSION

This nationwide, registry-based analysis of patients undergoing PCI for bypass graft lesions yielded several clinically relevant observations. First, approximately 30% of PCIs for bypass graft lesions were performed using a stentless PCI strategy with DCB in contemporary Japanese practice. Second, no statistically significant difference in 1-year clinical outcomes after PCI was observed between DES and DCB-only strategies within the constraints of this observational analysis (Central Illustration). In contrast, baseline patient risks, particularly chronic kidney disease and ACS at presentation, were significantly associated with an increased risk of MACE at 1 year. Taken together, these findings suggest that stentless PCI with DCB may be considered a treatment option for bypass graft lesions in selected patients, although further prospective studies are warranted.

TABLE 1 Baseline Characteristics of Patients Undergoing PCI for Bypass Graft Lesions

Characteristics	Overall Graft PCI (N = 610)			SVG PCI (n = 478)			AG PCI (n = 132)		
	DES (n = 426)	DCB (n = 184)	P Value	DES (n = 352)	DCB (n = 126)	P Value	DES (n = 74)	DCB (n = 58)	P Value
Age, y	73.7 ± 9.9	70.8 ± 11.1	0.002	74.1 ± 9.9	71.7 ± 10.6	0.03	71.8 ± 9.9	68.8 ± 11.9	0.12
Male	345/426 (81.0)	155/184 (84.2)	0.40	285/352 (81.0)	105/126 (83.3)	0.65	60/74 (81.1)	50/58 (86.2)	0.58
Previous history									
Hypertension	345/416 (82.9)	155/181 (85.6)	0.48	284/343 (82.8)	108/124 (87.1)	0.33	61/73 (83.6)	47/57 (82.5)	1.00
Diabetes	244/416 (58.7)	107/181 (59.1)	0.99	199/343 (58.0)	75/124 (60.5)	0.71	45/73 (61.6)	32/57 (56.1)	0.65
Hyperlipidemia	308/416 (74.0)	147/181 (81.2)	0.07	260/343 (75.8)	103/124 (83.1)	0.12	48/73 (65.8)	44/57 (77.2)	0.22
Current/recent smoker (within 1 year)	91/416 (21.9)	51/181 (28.2)	0.12	72/343 (21.0)	36/124 (29.0)	0.09	19/73 (26.0)	15/57 (26.3)	1.00
CKD including dialysis	182/416 (43.8)	76/181 (42.0)	0.76	147/343 (42.9)	57/124 (46.0)	0.62	35/73 (47.9)	19/57 (33.3)	0.13
Dialysis	67/416 (16.1)	31/181 (17.1)	0.85	52/343 (15.2)	24/124 (19.4)	0.35	15/73 (20.5)	7/57 (12.3)	0.31
Chronic lung disease	13/416 (3.1)	10/181 (5.5)	0.24	11/343 (3.2)	8/124 (6.5)	0.19	2/73 (2.7)	2/57 (3.5)	1.00
Peripheral arterial disease	69/416 (16.6)	39/181 (21.5)	0.18	56/343 (16.3)	27/124 (21.8)	0.22	13/73 (17.8)	12/57 (21.1)	0.81
Previous PCI	233/426 (54.7)	112/184 (60.9)	0.19	204/352 (58.0)	90/126 (71.4)	0.01	29/74 (39.2)	22/58 (37.9)	1.00
Previous myocardial infarction	212/422 (50.2)	89/183 (48.6)	0.78	169/350 (48.3)	62/125 (49.6)	0.88	43/72 (59.7)	27/58 (46.6)	0.19
Previous heart failure	127/424 (30.0)	52/181 (28.7)	0.84	106/350 (30.3)	36/124 (29.0)	0.88	21/74 (28.4)	16/57 (28.1)	1.00
Preprocedural characteristics									
Clinical presentations			0.051			0.35			0.11
Acute coronary syndrome	154/425 (36.2)	53/182 (29.1)		131/351 (37.3)	43/124 (34.7)		23/74 (31.1)	10/58 (17.2)	
STEMI	33/425 (7.8)	4/182 (2.2)		27/351 (7.7)	4/124 (3.2)		6/74 (8.1)	0/58 (0)	
NSTEMI	40/425 (9.4)	18/182 (9.9)		35/351 (10.0)	15/124 (12.1)		5/74 (6.8)	3/58 (5.2)	
Unstable angina	81/425 (19.1)	31/182 (17.0)		69/351 (19.7)	24/124 (19.4)		12/74 (16.2)	7/58 (12.1)	
Chronic coronary syndrome	271/425 (63.8)	129/182 (70.9)		220/351 (62.7)	81/124 (65.3)		51/74 (68.9)	48/58 (82.8)	
Cardiac arrest within 24 h	8/426 (1.9)	2/184 (1.1)	0.72	4/352 (1.1)	2/126 (1.6)	1.00	4/74 (5.4)	0/58 (0)	0.20
Acute heart failure within 24 h	20/426 (4.7)	5/184 (2.7)	0.36	16/352 (4.6)	4/126 (3.2)	0.69	4/74 (5.4)	1/58 (1.7)	0.52
Procedural characteristics									
Mechanical circulatory support during PCI									
IABP	13/426 (3.1)	4/184 (2.2)	0.74	10/352 (2.8)	4/126 (3.2)	1.00	3/74 (4.1)	0/58 (0)	0.34
ECMO	3/426 (0.7)	0/184 (0)	0.61	3/352 (0.9)	0/126 (0)	0.70	0/74 (0)	0/58 (0)	1.00
PVAD	0/426 (0)	0/184 (0)	1.00	0/352 (0)	0/126 (0)	1.00	0/74 (0)	0/58 (0)	1.00
In-hospital outcomes ^a									
Complications									
PCI-related myocardial infarction	1/426 (0.2)	2/184 (1.1)	0.45	0/352 (0)	2/126 (1.6)	0.12	1/74 (1.4)	0/58 (0)	1.00
Cardiac tamponade	0/426 (0)	0/184 (0)	1.00	0/352 (0)	0/126 (0)	1.00	0/74 (0)	0/58 (0)	1.00
Cardiogenic shock requiring mechanical and/or inotropic support	7/426 (1.6)	0/184 (0)	0.18	4/352 (1.1)	0/126 (0)	0.53	3/74 (4.1)	0/58 (0)	0.34
Stent thrombosis ^b	0/426 (0)	0/184 (0)	1.00	0/352 (0)	0/126 (0)	1.00	0/74 (0)	0/58 (0)	1.00
Requirement for emergency surgery	0/426 (0)	0/184 (0)	1.00	0/352 (0)	0/126 (0)	1.00	0/74 (0)	0/58 (0)	1.00
Bleeding requiring blood transfusion	1/426 (0.2)	1/184 (0.5)	1.00						
Access site bleeding	2/426 (0.5)	0/184 (0)	0.87	1/352 (0.3)	1/126 (0.8)	1.00	0/74 (0)	0/58 (0)	1.00
Non-access site bleeding	4/236 (0.9)	1/184 (0.5)	0.99	1/352 (0.3)	0/126 (0)	1.00	1/74 (1.4)	0/58 (0)	1.00

Values are mean ± SD or n/N (%) of patients. ^aPatients who died during hospitalization after PCI were excluded from this registry because follow-up after discharge could not be performed. ^bAs defined by the Academic Research Consortium.²⁷ The definition of stent thrombosis in the DCB group only applies in cases of in-stent restenosis, which differs from the current DCB Academic Research Consortium definition of thrombosis.²⁸

AG = arterial graft; CKD = chronic kidney disease; DCB = drug-coated balloon; DES = drug-eluting stent(s); ECMO = extracorporeal membrane oxygenation; IABP = intra-aortic balloon pump; NSTEMI = non-ST-segment elevation myocardial infarction; PCI = percutaneous coronary intervention; PVAD = percutaneous ventricular assist device; STEMI = ST-segment elevation myocardial infarction; SVG = saphenous vein graft.

INTERPRETATION OF THE MAIN FINDINGS AND POTENTIAL MECHANISMS SUPPORTING DCB USE.

The most clinically relevant finding of the present study is that the 1-year clinical outcomes did not significantly differ between DCB- and DES-based PCI

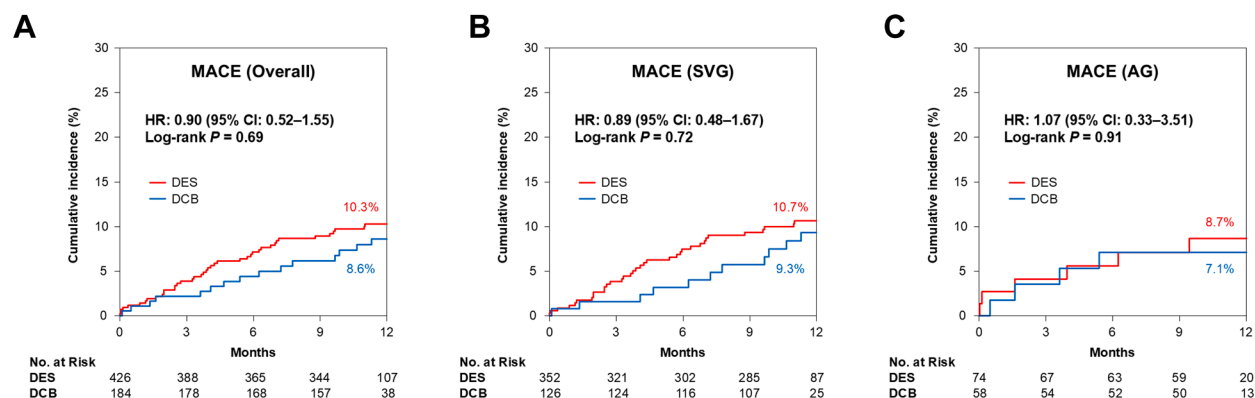
for bypass graft lesions. Furthermore, consistent results were found across SVG and AG subgroups, despite the known biological and structural differences between venous and arterial conduits.^{5,20} Although this analysis was not designed to explore

TABLE 2 Clinical Events at 1 Year After PCI for Bypass Graft Lesions

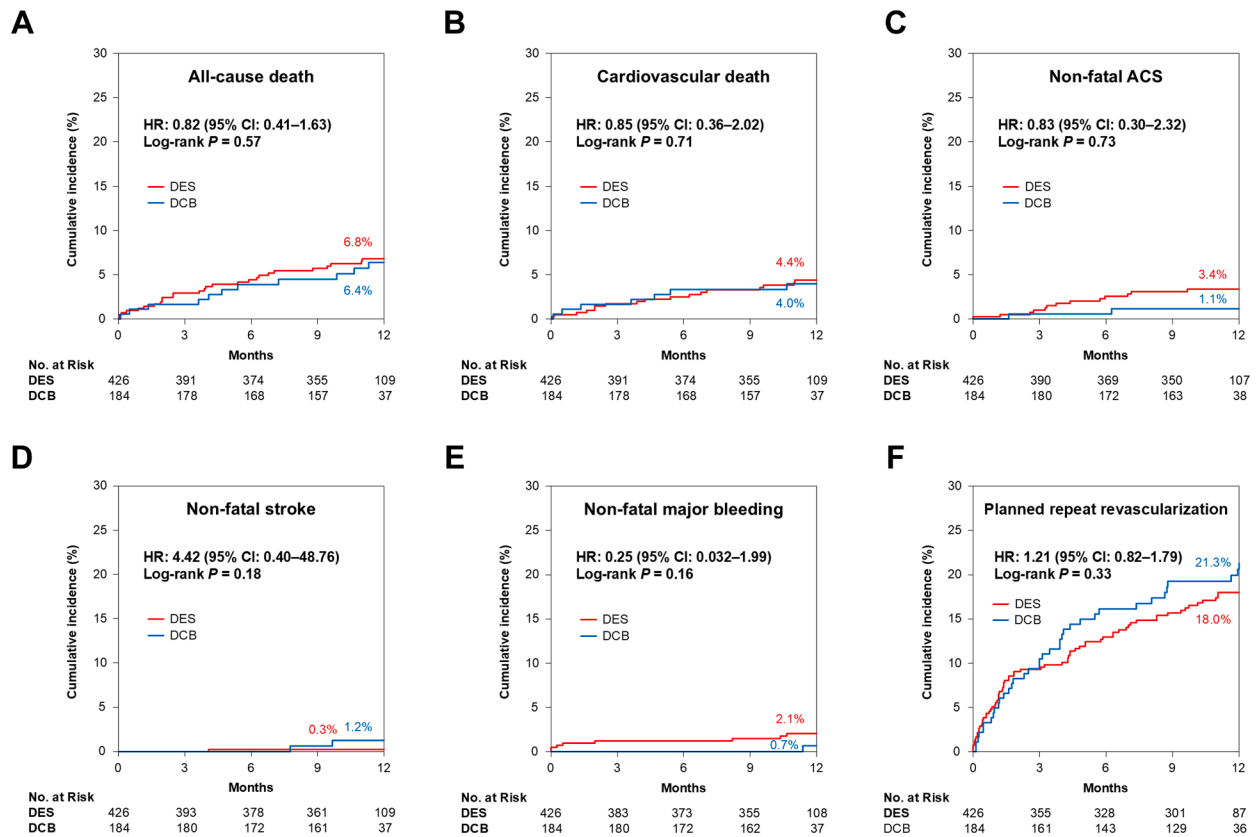
Outcome	Overall Graft PCI (N = 610)				SVG PCI (n = 478)				AG PCI (n = 132)			
	DES (n = 426)	DCB (n = 184)	HR (95% CI)	P Value	DES (n = 352)	DCB (n = 126)	HR (95% CI)	P Value	DES (n = 74)	DCB (n = 58)	HR (95% CI)	P Value
MACE ^a	41 (9.6)	15 (8.2)	0.90 (0.52-1.55)	0.69	35 (9.9)	11 (8.7)	0.89 (0.48-1.67)	0.72	6 (8.1)	4 (6.9)	1.07 (0.33-3.51)	0.91
All-cause death	27 (6.3)	11 (6.0)	0.82 (0.41-1.63)	0.57	21 (6.0)	8 (6.3)	0.90 (0.41-2.00)	0.80	6 (8.1)	3 (5.2)	0.61 (0.15-2.45)	0.49
Cardiovascular death	17 (4.0)	7 (3.8)	0.85 (0.36-2.02)	0.71	13 (3.7)	4 (3.2)	0.75 (0.25-2.25)	0.60	4 (5.4)	3 (5.2)	0.92 (0.20-4.09)	0.91
ACS death	4 (0.9)	0 (0)	Not estimable	-	3 (0.9)	0 (0)	Not estimable	-	1 (1.4)	0 (0)	Not estimable	-
Sudden cardiac death	2 (0.5)	3 (1.6)	3.33 (0.56-19.92)	0.19	2 (0.6)	1 (0.8)	1.33 (0.12-14.71)	0.81	0 (0)	2 (3.4)	Not estimable	-
Heart failure death	3 (0.7)	1 (0.5)	0.61 (0.07-5.46)	0.66	2 (0.6)	1 (0.8)	0.98 (0.10-9.46)	0.99	1 (1.4)	0 (0)	Not estimable	-
Stroke death	0 (0)	0 (0)	Not estimable	-	0 (0)	0 (0)	Not estimable	-	0 (0)	0 (0)	Not estimable	-
Major bleeding death	0 (0)	0 (0)	Not estimable	-	0 (0)	0 (0)	Not estimable	-	0 (0)	0 (0)	Not estimable	-
Noncardiovascular death	10 (2.3)	4 (2.2)	0.84 (0.27-2.63)	0.76	8 (2.3)	4 (3.2)	1.24 (0.38-4.03)	0.72	2 (2.7)	0 (0)	Not estimable	-
Nonfatal ACS	13 (3.1)	2 (1.1)	0.83 (0.30-2.32)	0.73	13 (3.7)	1 (0.8)	0.59 (0.17-2.06)	0.41	0 (0)	1 (1.7)	Not estimable	1.00
Nonfatal stroke	1 (0.2)	1 (0.5)	4.42 (0.40-48.76)	0.22	1 (0.3)	2 (1.6)	5.37 (0.49-59.22)	0.17	0 (0)	0 (0)	Not estimable	-
Nonfatal major bleeding	8 (1.9)	1 (0.5)	0.25 (0.03-1.99)	0.19	6 (1.7)	1 (0.8)	0.40 (0.05-3.21)	0.39	2 (2.7)	0 (0)	Not estimable	1.00
Acute heart failure	17 (4.0)	10 (5.4)	1.25 (0.58-2.71)	0.57	12 (3.4)	6 (4.8)	1.28 (0.48-3.36)	0.62	5 (6.8)	4 (6.9)	0.92 (0.25-3.43)	0.90
Planned repeat revascularization	70 (16.4)	37 (20.1)	1.21 (0.82-1.79)	0.33	59 (16.8)	27 (21.4)	1.28 (0.83-2.00)	0.27	11 (14.9)	10 (17.2)	1.11 (0.47-2.62)	0.81

Values are n (%) unless otherwise indicated. HRs are for the DCB group, with the DES group serving as the reference. ^aMACE was defined as a composite of all-cause death, nonfatal ACS, and nonfatal stroke within 1 year after PCI.

ACS = acute coronary syndrome; MACE = major adverse cardiovascular event(s); other abbreviations as in Table 1.

FIGURE 2 Kaplan-Meier Curves for 1-Year MACE Stratified by Treatment Strategy

The Kaplan-Meier curves show the cumulative incidence of major adverse cardiovascular events (MACEs) within 1 year after PCI according to treatment strategy. (A) Overall cohort, (B) SVG subgroup, and (C) AG subgroup. These figures compare time-to-event outcomes between the DES and DCB strategies across the overall population and by graft type. No statistically significant differences in 1-year MACE rates were observed between treatment strategies in any group. The percentages are the cumulative incidence of the event based on Kaplan-Meier estimates. Other abbreviations as in Figure 1.

FIGURE 3 Kaplan-Meier Curves for 1-Year Outcomes Stratified by Treatment Strategy

The Kaplan-Meier curves show the cumulative incidence of the following clinical outcomes within 1 year after PCI, according to treatment strategy. (A) All-cause death, (B) cardiovascular death, (C) nonfatal acute coronary syndrome (ACS), (D) nonfatal stroke, (E) nonfatal major bleeding, and (F) planned repeat revascularization. These figures compare time-to-event outcomes between DES and DCB strategies. No statistically significant differences were observed between treatment strategies in any group. The percentages are the cumulative incidence of the event based on Kaplan-Meier estimates. Other abbreviations as in [Figure 1](#).

mechanistic endpoints, several procedural and biological factors may explain why no statistically significant differences in clinical outcomes were observed between the 2 strategies.

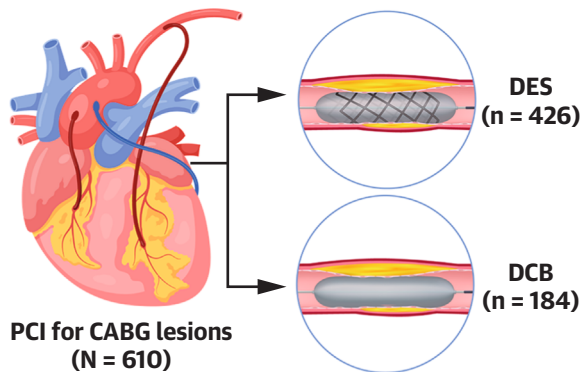
First, stent implantation in both SVGs and AGs, particularly near the anastomotic site, may compromise future revascularization options, increase metal burden, and lead to late complications, such as stent

TABLE 3 Multivariable Analysis for MACE Incidence at 1 Year

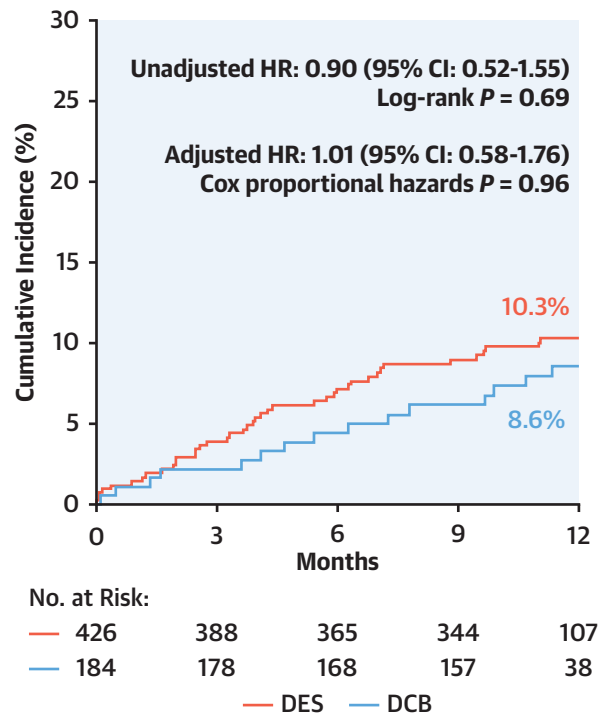
	Overall Graft PCI (N = 610)		SVG PCI (n = 478)		AG PCI (n = 132)	
	Adjusted HR (95% CI)	P Value	Adjusted HR (95% CI)	P Value	Adjusted HR (95% CI)	P Value
DCB (vs DES)	1.01 (0.58–1.76)	0.96	0.97 (0.52–1.84)	0.94	1.29 (0.38–4.41)	0.69
Age, per 1 year	1.00 (0.98–1.03)	0.90	1.00 (0.97–1.02)	0.79	1.02 (0.96–1.09)	0.51
Male	0.95 (0.50–1.81)	0.88	0.86 (0.43–1.73)	0.67	1.33 (0.27–6.55)	0.73
CKD including dialysis	2.07 (1.24–3.48)	0.006	2.33 (1.31–4.15)	0.004	1.35 (0.38–4.79)	0.64
ACS (vs CCS)	2.16 (1.30–3.60)	0.003	2.21 (1.26–3.88)	0.006	2.45 (0.66–9.08)	0.18

MACE was defined as a composite of all-cause death, nonfatal ACS, and nonfatal stroke within 1 year after PCI.

CCS = chronic coronary syndrome; other abbreviations as in [Tables 1 and 2](#).

CENTRAL ILLUSTRATION Comparison of Drug-Eluting Stents and Drug-Coated Balloons for Coronary Bypass Graft Lesions**1-Year Clinical Outcomes for DES vs DCB-Only Strategies in Patients Undergoing PCI for CABG Lesions****Study Design**

- Retrospective observational analysis using data from the J-PCI OUTCOME registry
- Primary endpoint: MACE (composite of all-cause death, ACS, and stroke) within 1 year after PCI
- Secondary endpoints: individual components of MACE, cardiovascular death, major bleeding, and planned repeat revascularization

1-Year MACE

- Stentless PCI with DCB was frequently employed in patients undergoing PCI for CABG lesions and was not associated with a significant difference in 1-year clinical outcomes compared with contemporary DES-based PCI (adjusted HR: 1.01, 95% CI: 0.58-1.76; P = 0.96).
- Results were consistent across saphenous vein and arterial graft subgroups.
- No statistically significant differences were observed in the secondary endpoints.

Aikawa T, et al. JACC Cardiovasc Interv. 2026;■(■):■-■.

Graft failure after coronary artery bypass grafting often requires percutaneous coronary intervention (PCI); however, the optimal PCI strategy for bypass graft lesions remains uncertain. Using data from a nationwide Japanese registry, we compared 1-year clinical outcomes between drug-eluting stent (DES) and drug-coated balloon (DCB)-only strategies in patients undergoing PCI for bypass graft lesions. Of the 610 PCIs, 184 (30.2%) were stentless PCIs with DCB. The 1-year major adverse cardiovascular event (MACE) rates did not differ significantly between the DES and DCB-only strategies, and the results were consistent across saphenous vein and arterial graft subgroups. These findings provide real-world evidence to inform contemporary PCI treatment strategies for bypass graft lesions. ACS = acute coronary syndrome; J-PCI OUTCOME = Japanese Percutaneous Coronary Intervention Outcome Registry.

fracture or neoatherosclerosis.⁵ A stentless strategy with DCB preserves vessel anatomy without permanent stent implantation and may be particularly advantageous for focal lesions, anastomotic disease, and patients with limited revascularization options.

Furthermore, using DCBs instead of stents may allow patients at high risk for bleeding to receive single antiplatelet therapy instead of prolonged dual antiplatelet therapy.^{21,22} These considerations support the potential role of DCB-based PCI in anatomically

complex bypass graft lesions, where minimizing permanent stent implantation is theoretically advantageous.

Second, SVG lesions are biologically distinct from native coronary lesions and are often characterized by friable atheroma and thrombotic components.^{5,20} Because stent implantation has been associated with an elevated risk of stent thrombosis and accelerated in-stent restenosis in patients undergoing SVG PCI compared with non-SVG PCI,²³ avoiding permanent stent implantation could theoretically minimize long-term vascular inflammation and subsequent adverse ischemic events. In the present study, similar rates of ACS and planned repeat revascularization after PCI were observed between the DES and DCB groups. These findings provide reassurance that avoiding permanent stent implantation was not associated with an apparent increase in ischemic events in this setting.

Third, late graft failure in AGs is relatively uncommon but can occur due to fibro-intimal proliferation or progressive atherosclerotic changes within the AGs.⁵ In this context, DCB-based PCI is biologically attractive because it enables local antiproliferative drug delivery without permanent stent implantation. Previous studies have reported that stentless PCI with DCBs for de novo lesions may facilitate positive vessel remodeling and late lumen enlargement after PCI,²⁴ supporting the biological plausibility of our findings. However, these mechanistic considerations remain speculative because detailed lesion-level information and serial angiographic follow-up data were unavailable in the present registry.

CLINICAL IMPLICATIONS. From a clinical perspective, the present study provides real-world data on contemporary treatment strategies for bypass graft lesions. DCB may be considered a treatment option for selected lesion subsets, such as focal lesions and anastomotic disease. A recent multicenter registry examined the feasibility and clinical outcomes of DCB-based PCI in 26 patients with bypass graft failure involving SVGs or AGs.⁸ During a median follow-up period of 75 months, the rate of adverse cardiovascular events remained relatively high, with 2 cases of nonfatal myocardial infarction and 2 cardiovascular deaths, despite modest target lesion revascularization rates of 11%. Taken together, these findings suggest that patient-related risk factors are more strongly associated with outcomes than the PCI device strategy itself. Therefore, careful patient selection and comprehensive secondary prevention

remain crucial for improving long-term outcomes after PCI for bypass graft lesions.^{5,6,20}

STUDY LIMITATIONS. This study has several limitations inherent to its observational, registry-based design. First, residual confounding factors cannot be excluded despite multivariable adjustment. Second, lesion-level analyses were not feasible because of the lack of detailed angiographic information, intravascular imaging data, and core laboratory adjudication in the J-PCI OUTCOME registry. Consequently, important lesion characteristics, such as reference vessel diameter, plaque morphology, lesion type (in-stent restenosis vs de novo lesion), and lesion location within the graft (ie, ostial, body, or distal anastomotic site), could not be evaluated. Third, the relatively small number of bypass-graft PCIs reflects the rarity of this subgroup in a national all-comer PCI registry, which may have increased the risk of type II error, particularly in subgroup analyses. Finally, clinical follow-up was limited to 1 year. The metal platform of the stent may demonstrate long-term efficacy against negative remodeling and elastic recoil. Therefore, the impact of the treatment strategy on long-term outcomes remains to be determined.

Several additional limitations relate to the device and procedural information. Only 1 paclitaxel-coated DCB was approved and used in Japan during the study period, which may limit the generalizability of our findings to other contemporary DCB platforms. However, a recent meta-analysis showed no significant differences in 1-year clinical outcomes between limus- and paclitaxel-coated DCBs.²⁵ The registry did not capture detailed information regarding DES type/generation or adjunctive procedural strategies, such as embolic protection devices, intravascular imaging guidance, lesion preparation techniques, and balloon to vessel diameter ratio. Therefore, our findings should be interpreted as a comparison of treatment strategies used in routine clinical practice, rather than as a direct comparison of specific device platforms. Furthermore, cases involving the concomitant use of DES and DCB were excluded because the intent of the procedure could not be determined. Some lesions initially planned for a DCB-only strategy may have required bailout DES implantation because of suboptimal angiographic results or procedural complications. Excluding these cases could introduce selection bias because they may represent real-world practice.

Propensity score matching or weighting is often used to reduce treatment selection bias in

observational studies. However, these approaches were not applied to the present analysis. This decision was based on methodological and data-related considerations. First, the registry lacked relevant lesion- and procedure-level variables, such as reference vessel diameter, lesion morphology, graft age, and intravascular imaging use. Consequently, propensity score models constructed using only patient-level clinical variables would likely fail to adequately capture the determinants of device selection and might not substantially improve on the multivariable Cox model used in this study. Second, the relatively small number of outcome events, particularly in the DCB group, precluded us from using more complex modeling approaches, which could have increased the risk of overfitting and reduced model stability.

Finally, several graft-specific factors could not be evaluated. The registry did not capture the interval between CABG and subsequent PCI, leaving the impact of graft aging on outcomes unknown. For SVGs, information regarding the harvesting technique was unavailable. SVGs obtained using the no-touch technique have been reported to be associated with improved graft patency compared with the conventional SVG harvesting approach.²⁶ In addition, AGs could not be classified by conduit type (ie, internal mammary artery, radial artery, or gastroepiploic artery).

Despite these limitations, the present study provides clinically relevant real-world evidence from a large, contemporary, nationwide cohort of patients undergoing PCI for bypass graft failure after CABG. This analysis directly compares DCB-based stentless PCI with DES-based strategies in a post-CABG population that is largely excluded from randomized trials,^{9,10} helping to address an important evidence gap. Future studies incorporating detailed graft- and lesion-level information and longer-term follow-up are needed to assess the long-term durability of these treatment strategies beyond 1 year after PCI because late restenosis and stent-related complications may occur during this time. Such efforts could help refine device selection and optimize revascularization in this complex clinical setting.

CONCLUSIONS

In this large nationwide registry, stentless PCI with DCB was frequently used in patients undergoing PCI

for bypass graft lesions and was not associated with a statistically significant difference in 1-year clinical outcomes compared with contemporary DES-based PCI strategies. These findings provide contemporary real-world data that may inform clinical decision-making in this high-risk population.

DATA AVAILABILITY This study was based on data that are not publicly available owing to a confidentiality agreement between the Japanese Association of Cardiovascular Intervention and Therapeutics and the data provider, the National Clinical Database (NCD) of Japan. The NCD will review reasonable requests for access to the minimum data set necessary to replicate the analysis and grant access consistent with applicable governance and confidentiality requirements. The statistical code is available from the corresponding author on reasonable request.

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PERSPECTIVES

WHAT IS KNOWN? Percutaneous coronary intervention (PCI) is frequently performed in clinical practice to treat bypass graft failure after coronary artery bypass grafting (CABG). Although the optimal device strategy remains debated, drug-eluting stents (DESs) are commonly used for graft lesions. PCI for diseased saphenous vein and arterial grafts presents unique procedural challenges, including distal embolization, restenosis, and variable vessel size. Evidence on stentless PCI with drug-coated balloons (DCBs) for bypass graft lesions is limited because randomized DCB trials often exclude patients with previous CABG or graft PCI.

WHAT IS NEW? A large, nationwide Japanese registry found that approximately one-third of PCIs for bypass graft failure were performed using a stentless PCI

strategy with DCB. No statistically significant differences in 1-year clinical outcomes were observed between DCB- and DES-based PCI strategies, with consistent findings across saphenous vein graft and arterial graft subgroups.

WHAT IS NEXT? Prospective studies and randomized trials are needed to better define the safety and effectiveness of DCB-based PCI for bypass graft lesions, as well as to identify optimal lesion- and patient-level selection criteria. Integrating intravascular imaging, conducting longer-term follow-up, and evaluating newer DCB platforms may further refine stentless treatment strategies for this complex post-CABG population.

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